

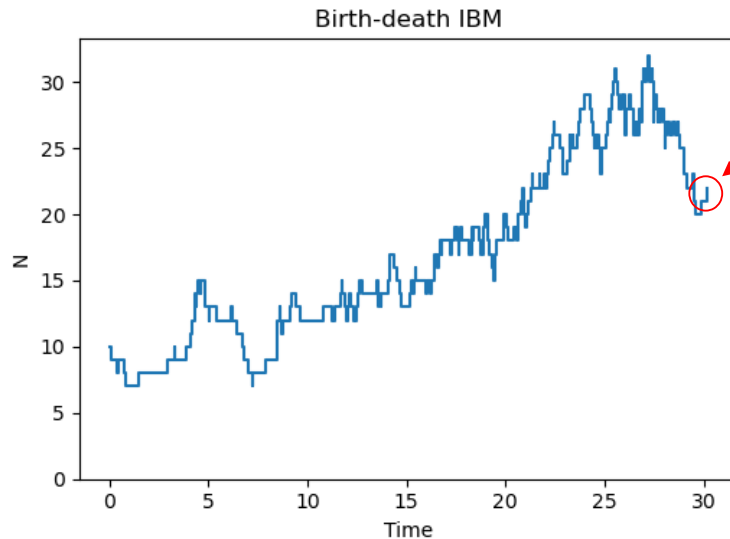
Announcements

- Updated slides and code for Tue

Today

- Making code faster continued
- Simulating birth-death IBM
 - distributions
- Monte Carlo error
- Emergence of deterministic model as the mean of the stochastic process

Code check & timings

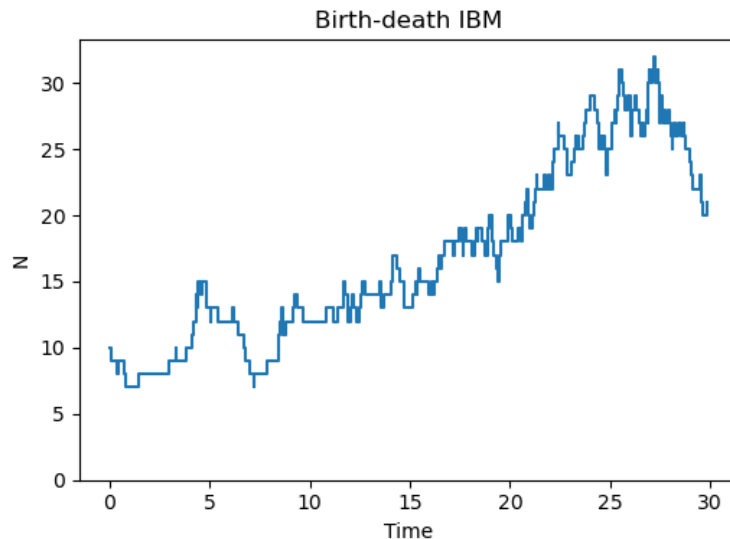


extra collected

Prototype

$N = 10$, $T = 0.068\text{s}$

$N = 1000$, $T = 172.3\text{s}$



Faster version

$N = 10$, $T = 0.011\text{ s}$

$N = 1000$, $T = 0.61\text{s}$

6X

280X

Monte Carlo error

With enough simulations (n), error $\rightarrow 0$

$$\text{CLT} \Rightarrow \bar{x} \sim \text{Normal}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

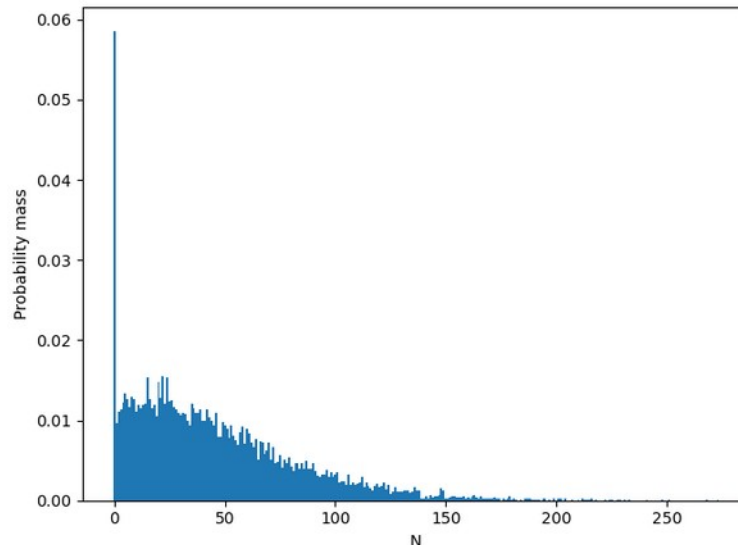
$$\text{MCSE} = \frac{\sigma}{\sqrt{n}}$$

$$95\% \text{ CI} \approx \bar{x} \pm 2 \times \text{MCSE}$$

Simulation study

PMF

$\Pr(N \mid t=30)$



sims = 10,000

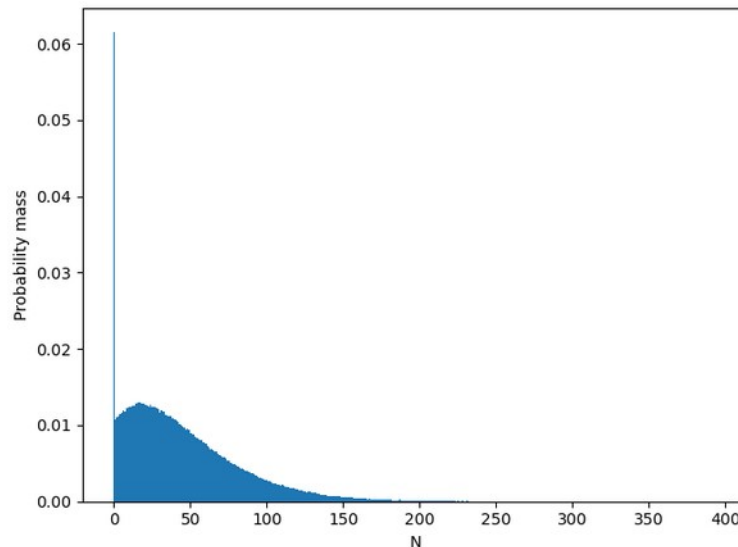
Expected N: 44.817

Observed $\bar{N}$: 45.404 $\pm$ 0.378

95% CI 44.6, 46.2

$P_{\text{extinct}} = 0.0586 \pm 0.0023$

95% CI 0.054, 0.063



sims = 1,000,000

Expected N: 44.817

Observed $\bar{N}$: 44.806 $\pm$ 0.037

95% CI 44.73, 44.88

$P_{\text{extinct}} = 0.0616$

Deterministic model emerges

